NETFLIX

DATA ANALYTICS

Streaming Trends, Content Strategy & Data-Driven Recommendations

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DTAN-500 | Indiana Wesleyan University | Prof. Lind Mary

CASE STUDY OVERVIEW

The Business Problem

Netflix faced rising subscriber churn and content saturation. With 200M+ subscribers globally, they needed smarter data-driven decisions to retain users and reduce content waste on titles nobody watched.

How Data Solved It

Netflix used viewer analytics — watch time, ratings, and genre preferences — to personalize recommendations, optimize content spending, and predict what new content to greenlight next.

Key Outcome

Data-driven personalization now influences over 80% of content watched on Netflix. It saves an estimated $1 billion annually through reduced churn and more targeted content investment decisions.

DATASET INTRODUCTION

DATASET SNAPSHOT

8,807
Total Titles

6,131
Movies

2,676
TV Shows

12
Columns of Data

2008–2021
Date Range

COLUMNS IN THE DATASET

- show_id — Unique ID for each title
- type — Movie or TV Show
- title — Name of the content
- director — Director's name
- country — Country of production
- date_added — When added to Netflix
- release_year — Original release year
- rating — Content rating (TV-MA, PG, etc.)
- duration — Minutes or number of seasons
- listed_in — Genre categories

Source: Kaggle — Netflix Movies and TV Shows (CC0 Public Domain)

DATA CLEANING & PREPARATION

01 Handled Missing Values

Removed rows where 'director' and 'country' fields were blank. Filled all empty 'date_added' cells with 'Unknown' so no rows were lost during trend analysis.

02 Removed Duplicates

Applied Excel's Remove Duplicates tool on the 'show_id' column. Result: zero true duplicates found — the dataset was already clean on unique title IDs.

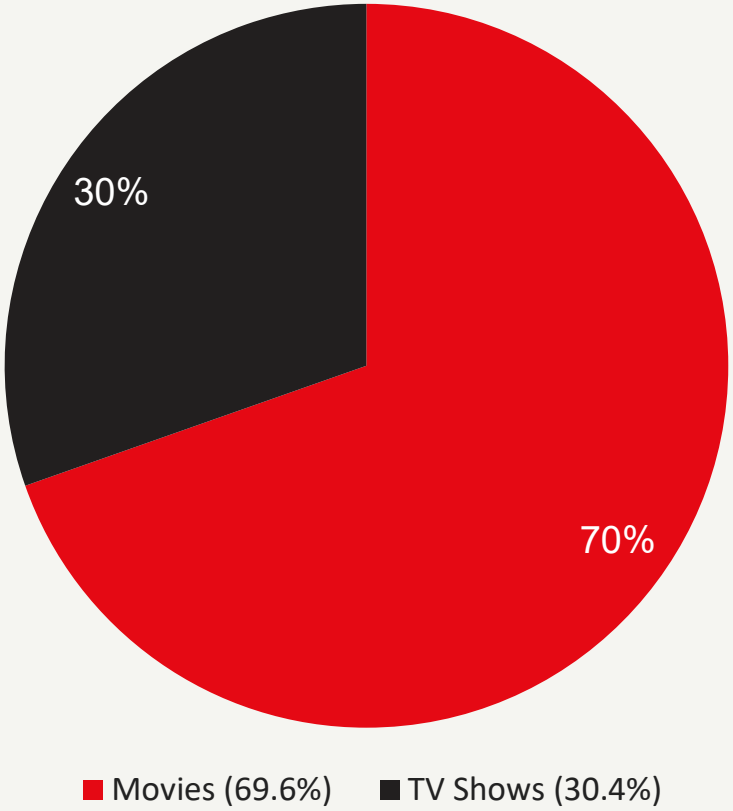
03 Standardized Date Formats

Reformatted 'date_added' into consistent MM/DD/YYYY format. Used Excel's YEAR() function to extract a new 'year_added' helper column for the content trend line chart.

04 Filtered & Trimmed Dataset

Reduced the full 8,807-row dataset to 1,000 rows for manageable Excel analysis. Removed 'description' and 'cast' columns not needed for the visualizations.

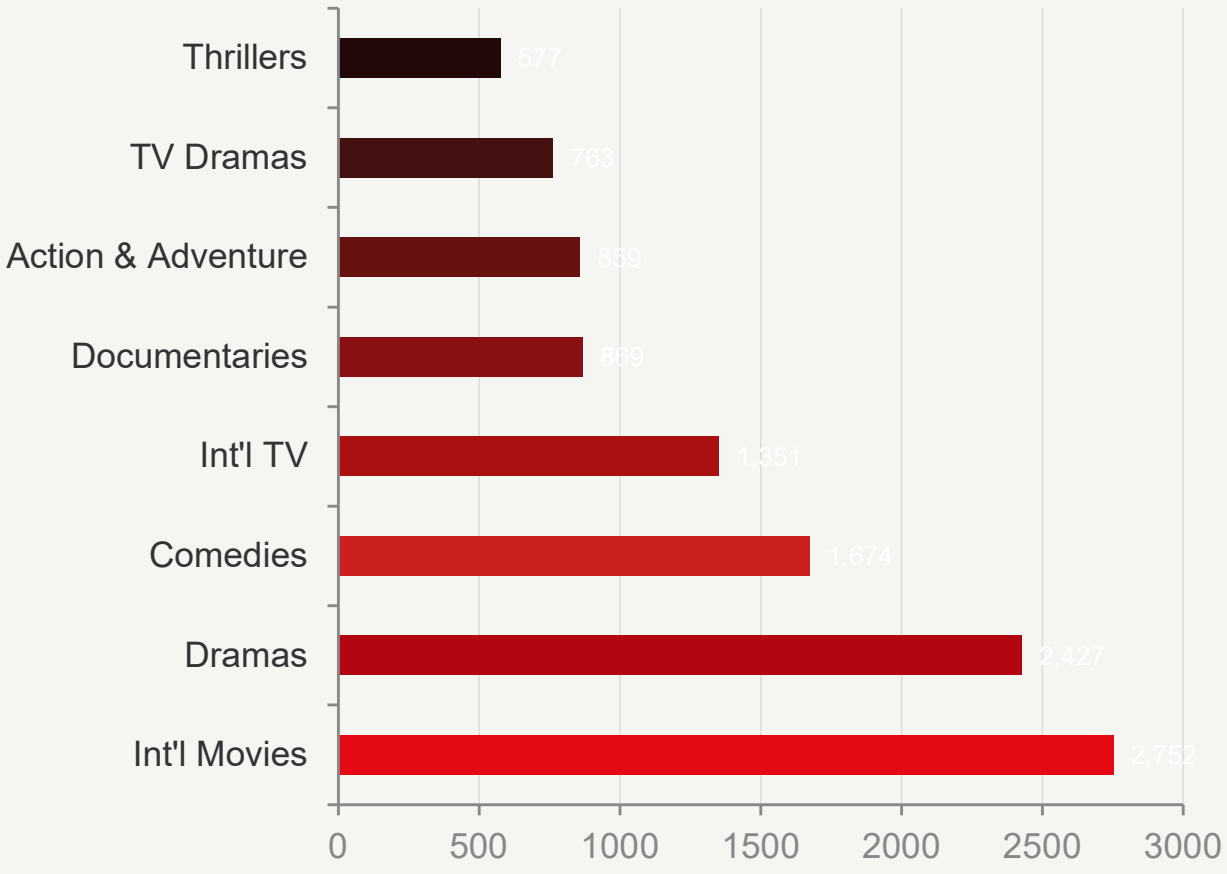
VISUALIZATION 1: Movies vs. TV Shows



KEY INSIGHTS

- Movies dominate Netflix's library at nearly 70% of all content.
- TV Shows, while smaller in count (30%), often drive longer engagement due to multi-episode format.
- This split suggests Netflix invests more in one-time movie licenses vs. multi-season series production.
- Opportunity: Growing TV Show catalog could improve subscriber retention metrics.

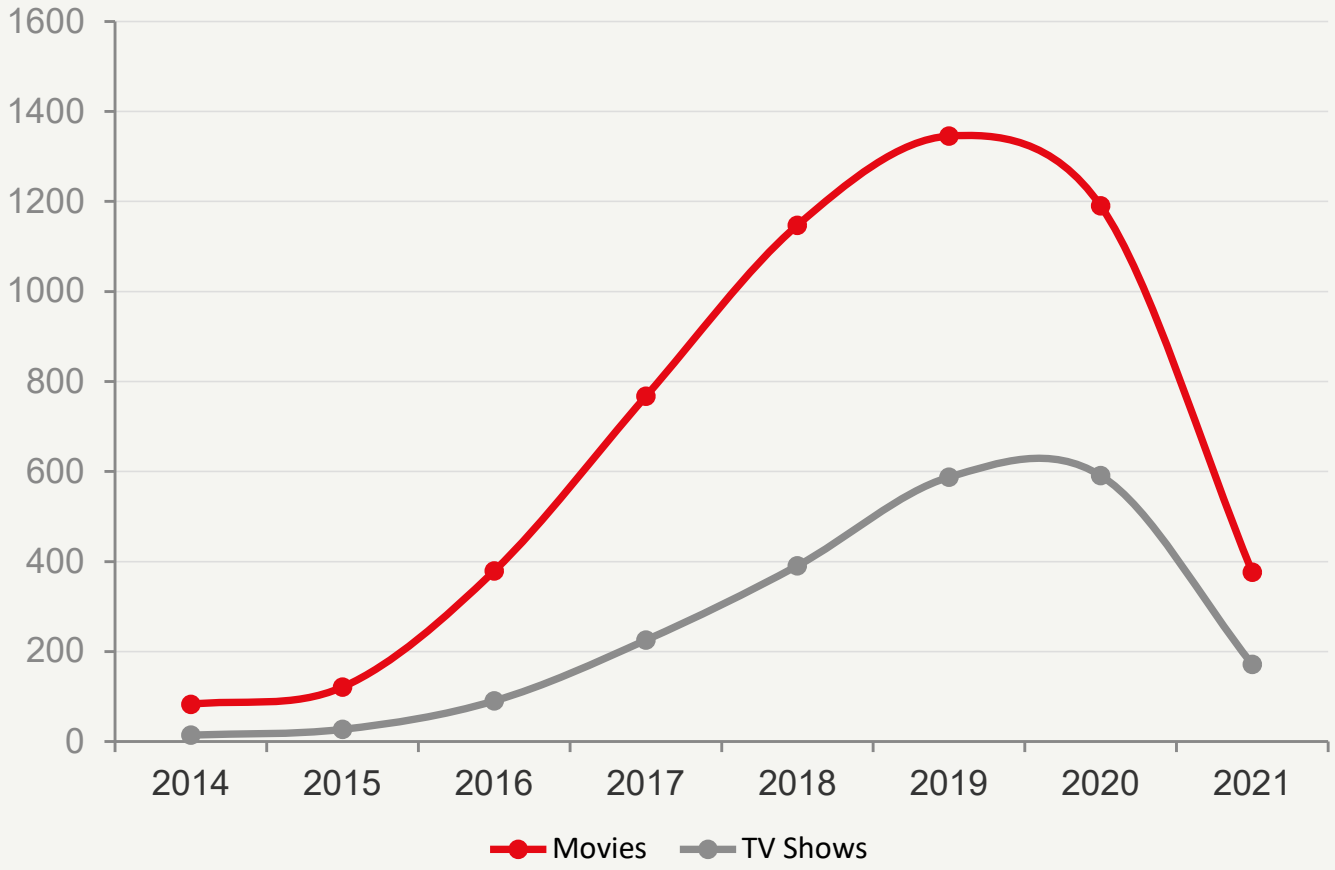
VISUALIZATION 2: Top 8 Content Genres



KEY INSIGHTS

- International Movies top the chart, showing Netflix's strong global content strategy.
- Dramas and Comedies rank 2nd and 3rd — core audience demand remains classic genres.
- International TV content is growing rapidly, driven by Korean dramas and Spanish series.
- Documentaries rank 5th, signaling growing demand for non-fiction content.

VISUALIZATION 3: Content Added to Netflix Per Year



KEY INSIGHTS

- Rapid content growth from 2016–2019 reflects Netflix's major investment phase.
- 2019 was peak year for new content additions.
- 2020–2021 shows a slowdown — likely due to COVID-19 production shutdowns.
- TV Show growth paralleled Movie growth, but at a smaller scale.

KEY FINDINGS

F1 Content is Movie-Heavy at 70%

Nearly 70% of Netflix's 8,807-title library consists of movies. While movies offer one-time viewing, TV shows drive longer watch sessions per subscriber — making them more powerful for reducing churn.

F2 International Content is the #1 Genre

International Movies (2,752 titles) and International TV Shows (1,351 titles) top all genre categories, confirming Netflix's deliberate strategy to invest in non-English global content.

F3 Content Growth Peaked in 2019

Netflix added its highest volume of new content in 2019. A sharp decline followed in 2020–2021, directly correlating with COVID-19 production shutdowns that halted filming worldwide.

F4 Mature Ratings Dominate the Library

TV-MA and R-rated titles make up the majority of Netflix's catalog, indicating the platform primarily targets adult viewers aged 18+, with limited offerings for children and families.

RECOMMENDATIONS

R1 Invest More in Original TV Series

TV shows generate longer watch sessions and stronger subscriber loyalty than movies. Netflix should shift its content budget toward original multi-season series to better compete with Disney+ and HBO Max.

R2 Deepen International Content Partnerships

International content is already the #1 genre by volume. Netflix should expand regional co-productions in Southeast Asia, India, and Latin America — the fastest-growing subscriber markets globally.

R3 Develop More Family-Friendly Content

With TV-MA dominating the library, Netflix is underserving families with children. Adding more TV-G and PG-rated originals would open a new subscriber segment and improve household plan retention.

R4 Data-Driven Content Greenlight Process

Netflix should use genre engagement data to decide which new titles to greenlight. Documentaries and Thrillers show strong audience demand at lower production cost — high-ROI categories worth prioritizing.

CONCLUSION

Data Analytics Drives Netflix's Competitive Advantage

This analysis of Netflix's 8,807-title dataset confirms that data is at the heart of every major content decision the company makes. From understanding which genres drive subscriber growth, to identifying when content investment peaks and declines, the data tells a clear and actionable story.

The four key findings — a movie-heavy library, dominance of international content, a COVID-driven content slowdown, and mature-rated titles dominating — all directly connect back to the Netflix Data Analytics case study. Organizations that embrace data gain a measurable edge in customer retention, content efficiency, and global market growth.

70%

Movies dominate the library

#1

International content leads all genres

80%

Personalization drives what users watch